

IBM BOB HACKATHON 2026

HarborLens

Explainable port congestion planning for the next 72 hours

Port operations copilot | AI track

Team: HarborLens Crew

Synthetic demo data. Human approval remains required.

Congestion appears after the useful action window

Port supervisors coordinate vessel arrivals, berth capacity, crane availability, and priority cargo across disconnected schedules and spreadsheets.

Fragmented inputs

Schedules show events, but not the operational trade-offs between vessels.

Cascading delays

A late vessel or reduced berth capacity can create queue pressure across the shift.

Priority cargo

Cold-chain and connection-sensitive loads require a defensible response before they wait offshore.

Challenge context: 2021 LA/Long Beach backlog exceeded 100 ships, with an estimated \$10B supply-chain cost.

Source: Industry Problem Statements 2026, Logistics and Ports L1

A transparent recommendation engine for the supervisor's shift

- 1

Ingest snapshot

Vessel schedules, berth capacity, cargo urgency, and disruption settings.
- 2

Score risk

Deterministic scoring exposes utilization, dwell, urgency, and scenario drivers.
- 3

Plan response

A compatible berth sequence and crane allocation create a 72-hour operating plan.
- 4

Brief with Bob

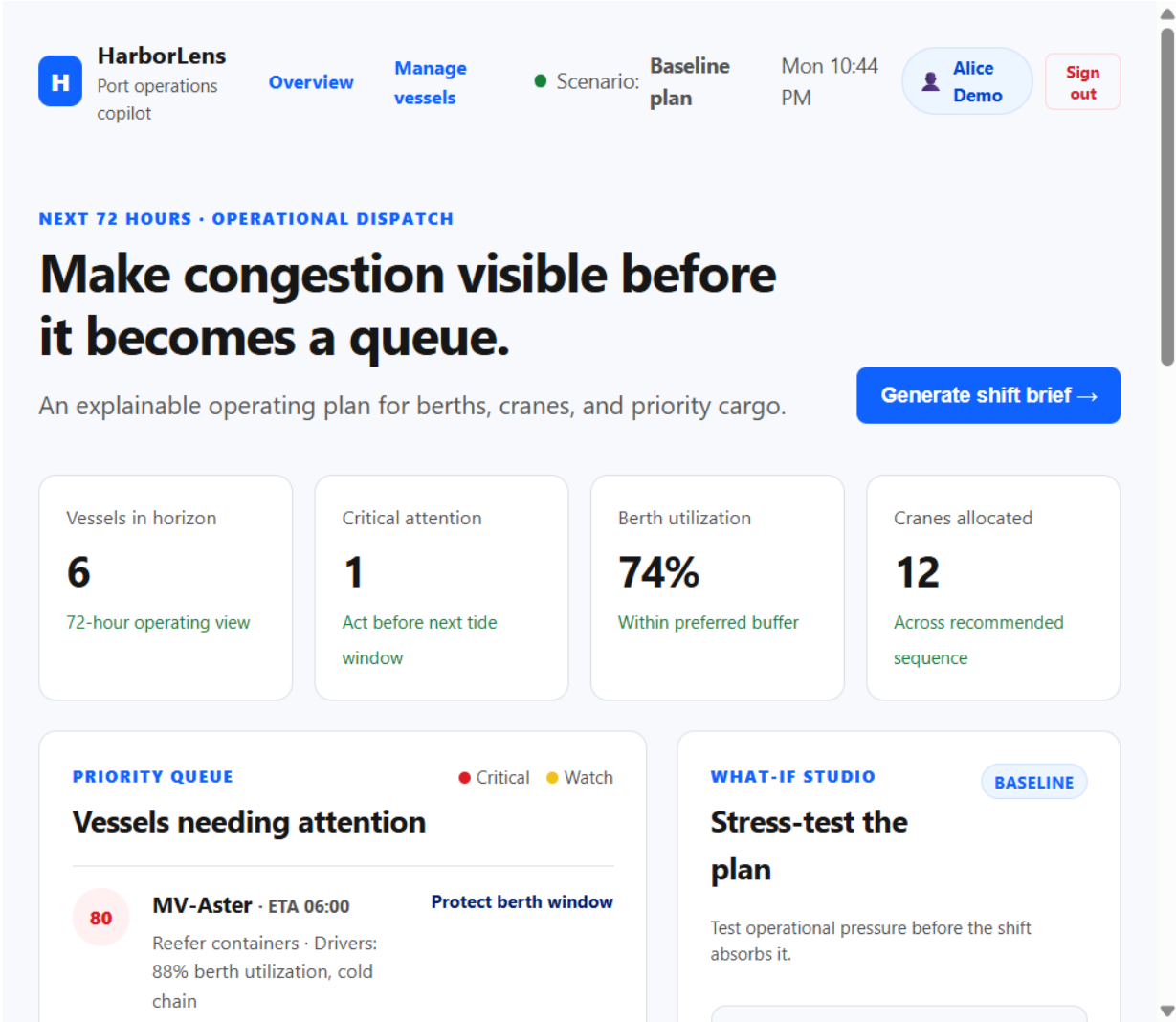
IBM Bob reviews grounded context and drafts an auditable BLUF shift brief.

The prototype never controls terminal systems. Supervisors review recommendations before any action.

Risk queue, what-if simulation, and grounded shift brief

Evidence in code

- Explainable risk drivers
- Berth and crane recommendation
- Weather, capacity, and ETA scenario
- Bob-ready structured briefing context



Each supervisor sees their own fleet, not a shared queue

Sign in

Register, login, or demo SSO issues a session token stored in the browser.

Own the data

GET/POST/PUT/DELETE /api/vessels require Bearer auth and filter by user_id.

Starter fleet

A new account receives a cloned 5-vessel snapshot so the demo is useful immediately.

Private edits

Adding MV-Alice-Only never appears on another operator's dashboard.

One FastAPI process: Neon Postgres + static UI

Frontend

dashboard.html / admin.html call /api/vessels with Authorization. Risk, berth plan, and what-if sliders stay client-side.

API + auth

FastAPI CRUD plus register/login/SSO-demo. Tokens are signed; not production IAM.

Neon Postgres

Remote SQLAlchemy store. No local Postgres. SQLite only if DATABASE_URL is unset.

IBM Bob skill: `src/bob-skills/port-operations.md` validates the same computed brief the UI shows.

Make congestion visible before it becomes a queue

- Supervisors get a ranked 72-hour picture with cited risk drivers.
- What-if weather, berth outage, and delay stay local — they do not rewrite the database.
- Bob produces a BLUF brief grounded in the current plan, not invented facts.
- Prototype only: synthetic data, human approval required, no terminal control.

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